

VII BAYES THEOREM

THE CONCEPT

$$P(A|B) = P(B|A) * P(A) / P(B)$$

PART	NAME	MEANING
$P(A B)$	Posterior	your updated belief about after seeing evidence B

PART	NAME	MEANING
$P(B A)$	Likelihood	How probable the evidence B is if A is true.
$P(A)$	Prior	Your belief about A before seeing any evidence.
$P(B)$	Evidence	Total probability of seeing B under all possibilities.

$$P(B) = P(B|A) * P(A) + P(B|A') * P(A')$$

NOTE: When condition is more even accurate tests produce mostly false positives.

- SPAM FILTER → a real spam filter applies Bayes across hundreds of words simultaneously.

- NAIVE BAYES → : Independence Assumption.

extends this to multiple features by assuming all features are conditionally independent given the class

$$P(\text{class} | \text{feature}_1, \text{feature}_2, \dots)$$

$$= \frac{P(\text{class}) * P(\text{feature}_1 | \text{class}) * P(\text{feature}_2 | \text{class})}{P(\text{feature}_1, \text{feature}_2, \dots)}$$

MAXIMUM LIKELIHOOD ESTIMATION

$$P(\text{"free"} | \text{spam}) = \frac{\text{no. of spam emails containing "free"}}{\text{total spam emails}}$$

chooses the parameter values that makes the observed data most likely

laplace smoothing

$$P(\text{word} | \text{class}) = \frac{(\text{count}(\text{word}, \text{class}) + 1)}{(\text{total words in class} + \text{total words})}$$

Maximum APOSTERIORI (MAP)

~~MAP~~ MLE: what params maximize $P(\text{data} | \text{params})$

MAP: what params maximize $P(\text{params} | \text{data})$

$$P(\text{params} | \text{data}) \propto P(\text{data} | \text{params}) * P(\text{params})$$

MAP adds a prior over the params themselves

small params $\rightarrow$ encode as prior so large values would be penalized

Identical to L_2 regularization

MLE	$P(\text{data} \text{params})$	Irregular and
MAP	$P(\text{data} \text{params}) * P(\text{params})$	4/22 regular

BAYESIAN VS FREQUENTIST: THE PRACTICAL DIFF

Parameters as fixed unknown distribution.

Params as fixed unknowns.

Given what I observed what do I believe.

If I repeat this many times what would happen.

Output { Distribution

Point estimate.

Uncertainty { Credible Intervals

Confidence Intervals

small data prior acts as regularization

Can Overfit

Computation { requires sampling

Usually fast.

* Why Bayesian Thinking matters for ML

- Priors are regularization.

- Posterior are uncertainty.

- Bayesian Updates are online learning.

CONJUGATE PRIORS :-

When prior and posterior belongs to same ~~class~~ family of distribution is called conjugate.
(family of distribution)

→ A/B website testing button.
if conversion $\rightarrow$ 20%.

$$\text{Beta}(\alpha=2, \beta=8) \Rightarrow \frac{\alpha}{\alpha+\beta} = 0.2$$

if on a new day we have
7 conversion out of 20 visitors.

$$\text{Beta}(\alpha=\alpha+7, \beta=\beta+13) \Rightarrow 0.30$$

SEQUENTIAL BAYESIAN UPDATING

Today's posterior becomes tomorrow's prior
this is how real systems learn
incrementally without supplanting
all historical data.

↓
foundation of online learning in
production ML systems.